

⚡ API Quick Reference ⚡

💠 HTTP Verbs

- └  **GET** : Retrieve data from the server
- └  **POST** : Send data to the server to create a resource
- └  **PUT** : Send data to the server to update a resource
- └  **PATCH** : Send data to the server to update a resource partially
- └  **DELETE** : Delete a resource from the server
- └  **TRACE** : Returns the full HTTP request received by the server for debugging and diagnostic purposes
- └  **OPTIONS** : Returns the HTTP methods supported by the server for the requested URL
- └  **CONNECT** : Converts the request connection to a transparent TCP/IP tunnel for secure communication
- └  **PURGE** : Invalidates a cached resource
- └  **LOCK** : Locks the resource for exclusive use by the client
- └  **UNLOCK** : Unlocks the resource previously locked by the client
- └  **MKCOL** : Creates a new collection resource
- └  **COPY** : Copies the resource identified by the Request-URI to the destination URI.

💠 HTTP Status Codes

- └  **1xx** : Informational
- └  **2xx** : Success
- └  **3xx** : Redirection
- └  **4xx** : Client Errors
- └  **5xx** : Server Errors

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💠 Response Headers

- └  **Content-Type** : Specifies the MIME type of the data in the response body
- └  **Content-Length** : Specifies the length of the response body in bytes
- └  **Cache-Control** : Specifies the caching behavior of the response
- └  **Location** : Specifies the URI of a resource that can be used to retrieve the requested resource
- └  **Server** : Specifies the name and version of the server software that generated the response
- └  **Access-Control-Allow-Origin** : Specifies which origins are allowed to access the resource
- └  **Set-Cookie** : Specifies a cookie that should be stored by the client and sent back to the server with future requests
- └  **Expires** : Specifies the date and time after which the response is considered stale
- └  **Last-Modified** : Specifies the date and time the resource was last modified.

💠 API Design

- └  **REST** : Representational State Transfer, a design pattern for building web services
- └  **SOAP** : Simple Object Access Protocol, a messaging protocol for exchanging structured data
- └  **GraphQL** : A query language and runtime for building APIs
- └  **API Gateway** : A service that manages, protects, and scales APIs

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🔹 API Architectures

- └ 🏠 **SOA** : Service-Oriented Architecture, an architectural style for building distributed systems
- └ 🏠 **Microservices** : An architectural style for building complex applications as a suite of small, independent services
- └ 🏠 **Serverless** : A cloud computing execution model where the cloud provider manages the infrastructure and automatically allocates resources as needed
- └ 🏠 **Event-Driven** : An architectural style where the flow of data between components is triggered by events
- └ 🏠 **RESTful API** : An architectural style that uses HTTP requests to GET, POST, PUT, and DELETE data.

🔹 API Design Patterns

- └ 🌸 **Adapter Pattern** : A pattern that converts the interface of a class into another interface that clients expect
- └ 🌸 **Decorator Pattern** : A pattern that adds behavior to an individual object dynamically
- └ 🌸 **Proxy Pattern** : A pattern that provides a surrogate or placeholder for another object to control access to it
- └ 🌸 **Chain of Responsibility Pattern** : A pattern that delegates commands to a chain of processing objects
- └ 🌸 **Observer Pattern** : A pattern that defines a one-to-many dependency between objects so that when one object changes state, all its dependents are notified and updated automatically.

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💠 API Security

- └ 🛡️ **OAuth** : An open standard for authorization used for protecting APIs
- └ 🛡️ **JWT** : JSON Web Tokens, a standard for securely transmitting information between parties as a JSON object
- └ 🛡️ **SSL/TLS** : Secure Sockets Layer/Transport Layer Security, a protocol for establishing a secure connection between a client and a server
- └ 🛡️ **API Key** : A secret token used to authenticate API requests
- └ 🛡️ **Rate Limiting** : A technique used to limit the number of requests that can be made to an API over a specific period of time
- └ 🛡️ **OpenID Connect** : An authentication layer built on top of OAuth that allows users to be authenticated across multiple domains
- └ 🛡️ **Cross-Origin Resource Sharing (CORS)** : A mechanism that allows many resources (e.g., fonts, JavaScript, etc.) on a web page to be requested from another domain outside the domain from which the resource originated

💠 API Testing

- └ 🛠️ **Postman** : A popular tool for testing and debugging APIs
- └ 🛠️ **SoapUI** : A tool for testing SOAP and REST web services
- └ 🛠️ **Swagger** : A tool for designing, building, and testing APIs
- └ 🛠️ **JMeter** : A tool for testing the performance of APIs
- └ 🛠️ **TestRail** : A test management tool for planning, executing, and tracking API tests
- └ 🛠️ **Dredd** : A command-line tool for testing API documentation against its backend implementation
- └ 🛠️ **REST Assured** : A Java-based library for testing RESTful APIs

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- └  **Karate DSL** : A testing framework for API testing using Gherkin syntax
- └  **HttpMaster** : A tool for testing and debugging APIs
- └  **Assertible** : A tool for testing and monitoring APIs with automated tests.

🔹 API Development

- └  **Node.js** : A JavaScript runtime for building server-side applications
- └  **Express** : A popular framework for building web applications and APIs with Node.js
- └  **Django** : A Python web framework for building web applications and APIs
- └  **Flask** : A lightweight Python web framework for building web applications and APIs
- └  **Spring** : A Java framework for building enterprise-level web applications and APIs
- └  **Swagger Editor** : A tool for designing and documenting APIs using the OpenAPI specification
- └  **Postman** : A tool for testing and debugging APIs
- └  **Insomnia** : A tool for designing, testing, and debugging APIs
- └  **Paw** : A tool for designing and testing APIs on Mac OS
- └  **API Blueprint** : A high-level API description language for building RESTful APIs.

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💠 API Implementation Platforms

- └  **Firebase** : A mobile and web application development platform developed by Google
- └  **Backendless** : A mobile and web application development platform that allows developers to build and deploy applications without backend coding
- └  **Parse Server** : An open-source version of the Parse backend that can be deployed to any infrastructure
- └  **Amazon API Gateway** : A fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs
- └  **Microsoft Azure API Management** : A fully managed service that enables users to publish, secure, transform, maintain, and monitor APIs.

💠 API Performance

- └  **Caching** : A technique for improving API performance by storing responses in a cache
- └  **Throttling** : A technique for limiting the rate of requests to an API to prevent overload
- └  **Load Balancing** : A technique for distributing traffic evenly across multiple servers to improve API performance
- └  **Content Delivery Network (CDN)** : A distributed system of servers that delivers content to users based on their geographic location to improve API performance
- └  **Edge Computing** : A computing paradigm that brings computation and data storage closer to the location where it is needed to reduce latency and improve API performance.

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💠 API Monitoring

- └ 🕵️ **Pingdom** : A tool for monitoring the uptime and performance of APIs
- └ 🕵️ **New Relic** : A tool for monitoring the performance of APIs and other web applications
- └ 🕵️ **Datadog** : A monitoring and analytics platform for cloud-scale applications and APIs
- └ 🕵️ **Sumo Logic** : A cloud-based log management and analytics platform for APIs and other applications
- └ 🕵️ **Loggly** : A cloud-based log management platform for monitoring APIs and other applications

💠 API Standards

- └ 🖋️ **JSON API** : A specification for building APIs that use JSON as the data format
- └ 🖋️ **HAL** : Hypertext Application Language, a standard for building hypermedia-driven APIs
- └ 🖋️ **JSON-LD** : A format for representing linked data on the web
- └ 🖋️ **OData** : Open Data Protocol, a standard for building and consuming RESTful APIs
- └ 🖋️ **AsyncAPI** : A specification for building event-driven APIs.

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🔹 API Standards Organizations

- └ 🌐 **W3C** : The World Wide Web Consortium, an international community that develops web standards
- └ 🌐 **IETF** : The Internet Engineering Task Force, an open standards organization that develops and promotes Internet standards
- └ 🌐 **OASIS** : Organization for the Advancement of Structured Information Standards, a nonprofit consortium that drives the development, convergence, and adoption of open standards for the global information society
- └ 🌐 **RESTful API Modeling Language (RAML)** : A YAML-based language for describing RESTful APIs developed by MuleSoft
- └ 🌐 **JSON API** : A specification for building APIs that use JSON as the data format.

🔹 API Infrastructure

- └ 📦 **Kubernetes** : An open-source platform for managing containerized workloads and services
- └ 📦 **OpenShift** : A container application platform that builds on top of Kubernetes
- └ 📦 **Docker Swarm** : A native clustering and orchestration solution for Docker
- └ 📦 **Consul** : A service mesh solution that provides service discovery, configuration, and segmentation capabilities
- └ 📦 **Istio** : A service mesh solution that provides traffic management, security, and observability capabilities.

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💠 API Governance

- └ 📄 **API Management** : The process of creating, publishing, and monitoring APIs in a secure and scalable way
- └ 📄 **API Monetization** : The process of generating revenue from APIs by charging developers for usage
- └ 📄 **API Versioning** : The process of managing changes to APIs over time
- └ 📄 **API Analytics** : The process of collecting and analyzing data on API usage and performance
- └ 📄 **API Gateway** : A service that manages, protects, and scales APIs.

💠 API Documentation

- └ 📖 **OpenAPI** : A specification for building APIs in YAML or JSON format
- └ 📖 **API Blueprint** : A high-level API description language for building RESTful APIs
- └ 📖 **RAML** : A YAML-based language for describing RESTful APIs
- └ 📖 **Swagger UI** : A tool for visualizing and interacting with APIs that have been described using the OpenAPI specification
- └ 📖 **Slate** : A tool for generating beautiful, responsive API documentation.

💠 API Deployment

- └ 🚀 **Heroku** : A cloud platform for deploying, managing, and scaling web applications and APIs
- └ 🚀 **AWS Elastic Beanstalk** : A service for deploying and scaling web applications and APIs on AWS

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└  **Azure App Service** : A service for deploying and scaling web applications and APIs on Azure

└  **Google App Engine** : A service for deploying and scaling web applications and APIs on GCP

└  **Docker** : A containerization platform used for packaging and deploying applications

└  **AWS Lambda** : A serverless compute service for running code in response to events

└  **Azure Functions** : A serverless compute service for running code in response to events

└  **Google Cloud Functions** : A serverless compute service for running code in response to events

└  **Netlify** : A cloud platform for deploying and managing static websites and APIs

└  **Vercel** : A cloud platform for deploying and managing static websites and APIs

💎 API Security

└  **OAuth** : An open standard for authorization used by many social media platforms and APIs

└  **OpenID Connect** : An authentication layer built on top of OAuth that allows users to be authenticated across multiple domains

└  **JSON Web Tokens (JWT)** : A method for representing claims securely between two parties

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└  **Cross-Origin Resource Sharing (CORS)** : A mechanism that allows many resources (e.g., fonts, JavaScript, etc.) on a web page to be requested from another domain outside the domain from which the resource originated

└  **API Keys** : A secret token that identifies an API client to the server and allows the client to access resources.

💠 API Best Practices

└  **Versioning** : A technique for managing changes to APIs over time

└  **Pagination** : A technique for breaking up large API responses into smaller, more manageable chunks

└  **Caching** : A technique for improving API performance by storing responses in a cache

└  **Error Handling** : A technique for returning meaningful error messages to API clients

└  **HATEOAS** : Hypermedia as the Engine of Application State, a constraint of RESTful APIs that requires the API to provide links to related resources

💠 API Tutorials

└  Getting Started with RESTful APIs by Tania Rascia

└  API Design Best Practices by Martin Fowler

└  Testing RESTful Web Services Made Easy Using the REST Assured Framework by Dinesh Rajput

└  API Gateway Concepts and Options by AWS

└  Building Secure APIs by Auth0

└  RESTful API Designing guidelines – The best practices by Mahesh Haldar

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💠 API Guides

- └ 📖 REST API Tutorial by Guru99
- └ 📖 A Beginner's Guide to HTTP and REST by Linode
- └ 📖 REST API Design: Resource Modeling by Oracle
- └ 📖 API Security Best Practices by Google Cloud
- └ 📖 API Governance Handbook by WS02.

💠 API Tools

- └ 🛠 **API Studio** : A web-based IDE for designing and testing APIs
- └ 🛠 **Stoplight** : A collaborative platform for designing, documenting, and testing APIs
- └ 🛠 **Apigee** : A full lifecycle API management platform that allows developers to design, secure, deploy, and analyze APIs
- └ 🛠 **Azure API Management** : A fully managed service that enables users to publish, secure, transform, maintain, and monitor APIs
- └ 🛠 **Postman Learning Center** : A hub for learning how to use Postman to design, develop, and test APIs.